

Consider the data set and perform the Apriori Algorithm and FP algorithm support:3 and confidence=50%

Consider the market basket transactions shown in the above table.

(a) What is the maximum number of association rules that can be extracted from this data (including rules that have zero support)?

(b) What is the maximum size of frequent itemsets that can be extracted (assuming minsup > 0)?

| Transaction ID | Items Bought                   |
|----------------|--------------------------------|
| 1              | {Milk, Beer, Diapers}          |
| 2              | {Bread, Butter, Milk}          |
| 3              | {Milk, Diapers, Cookies}       |
| 4              | {Bread, Butter, Cookies}       |
| 5              | {Beer, Cookies, Diapers}       |
| 6              | {Milk, Diapers, Bread, Butter} |
| 7              | {Bread, Butter, Diapers}       |
| 8              | {Beer, Diapers}                |
| 9              | {Milk, Diapers, Bread, Butter} |
| 10             | {Beer, Cookies}                |

Associator

Choose
Apriori -N 10 -T 0 -C 0.9 -D 0.05 -U 1.0 -M 0.1 -S -1.0 -c -1

Start
Stop

Result list (right-click for ...)

09:14:38 - Apriori

Associator output

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Apriori
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Minimum support: 0.35 (3 instances)
Minimum metric <confidence>: 0.9
Number of cycles performed: 13

Generated sets of large itemsets:

Size of set of large itemsets L(1): 6

Size of set of large itemsets L(2): 5

Size of set of large itemsets L(3): 1

Best rules found:

1. BEER=BUTTER 3 ==> I%2MILK=BREAD 3    <conf:(1)> lift:(3) lev:(0.22) [2] conv:(2)
2. I%2MILK=BREAD 3 ==> BEER=BUTTER 3    <conf:(1)> lift:(3) lev:(0.22) [2] conv:(2)
3. I%2MILK=BREAD 3 ==> NULL=NULL 3      <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
4. I%2MILK=MILK 3 ==> BEER=DIAPERS 3    <conf:(1)> lift:(2.25) lev:(0.19) [1] conv:(1.67)
5. I%2MILK=BEER 3 ==> NULL=NULL 3      <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
6. BEER=BUTTER 3 ==> NULL=NULL 3      <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
7. BEER=BUTTER NULL=NULL 3 ==> I%2MILK=BREAD 3    <conf:(1)> lift:(3) lev:(0.22) [2] conv:(2)
8. I%2MILK=BREAD NULL=NULL 3 ==> BEER=BUTTER 3    <conf:(1)> lift:(3) lev:(0.22) [2] conv:(2)
9. I%2MILK=BREAD BEER=BUTTER 3 ==> NULL=NULL 3    <conf:(1)> lift:(1.29) lev:(0.07) [0] conv:(0.67)
10. BEER=BUTTER 3 ==> I%2MILK=BREAD NULL=NULL 3    <conf:(1)> lift:(3) lev:(0.22) [2] conv:(2)

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